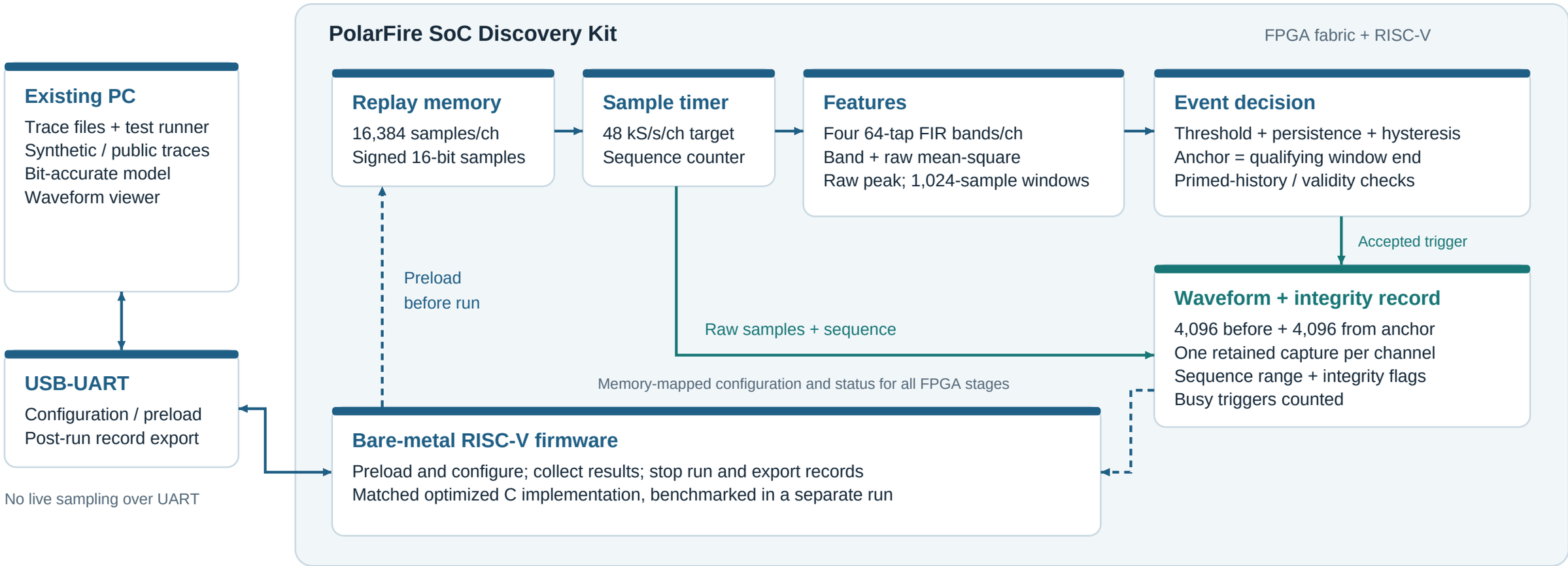
VibeTrace

Proposed system block diagram

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PROPOSED OPERATING TARGETS

Start with 1 channel at 48 ksample/s; attempt 4 channels at that rate if timing and placement permit.
4-channel sample storage: 128 KiB replay + 64 KiB capture = 192 KiB, plus filter state and queues.

All rates, capacities and performance goals are proposed and unmeasured. Baseline uses memory replay; no live sensor or ADC is required.
After one capture per channel is complete, records freeze while feature processing continues; subsequent busy triggers are counted.
Incomplete history, filter warm-up, clipping, arithmetic saturation and FIFO overflow must be reported explicitly.